

ESAT PRACTICE TEST SERIES

MATHS 2 + BIOLOGY

LEVEL 0 • EASY

Maths 2 + Biology

Mathematics 1 + Mathematics 2 + Biology

Foundation-focused ESAT practice with more direct setup and shorter solution chains.

MODULE

Mathematics 1

MODULE

Mathematics 2

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Maths 2 + Biology · Level 0 Easy

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 1 · preserved Mathematics 1 module

Q1 G	Q2 C	Q3 B	Q4 B	Q5 E
Q6 G	Q7 D	Q8 C	Q9 B	Q10 C
Q11 A	Q12 D	Q13 B	Q14 A	Q15 D
Q16 C	Q17 B	Q18 D	Q19 C	Q20 E
Q21 E	Q22 C	Q23 E	Q24 D	Q25 E
Q26 C	Q27 C			

Mathematics 2

Engineering Mock Test 1 · preserved Mathematics 2 module

Q1 A	Q2 F	Q3 D	Q4 A	Q5 D
Q6 D	Q7 F	Q8 B	Q9 D	Q10 D
Q11 B	Q12 E	Q13 E	Q14 B	Q15 C
Q16 A	Q17 D	Q18 A	Q19 C	Q20 B
Q21 A	Q22 A	Q23 C	Q24 C	Q25 D
Q26 C	Q27 D			

Biology

Preserved Biology Easy source set

Q1 C	Q2 A	Q3 H	Q4 C	Q5 E
Q6 E	Q7 C	Q8 D	Q9 B	Q10 B
Q11 E	Q12 G	Q13 C	Q14 D	Q15 B
Q16 B	Q17 E	Q18 E	Q19 E	Q20 E
Q21 D	Q22 A	Q23 B	Q24 C	Q25 B
Q26 H	Q27 B			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 1 - Mathematics 1

Answer **G****QUESTION 1**

If the 5-digit number $1X2X3$ is divisible by 9, what is X ?

A 0**B** 1**C** 2**D** 3**E** 4**F** 5**G** 6**WORKED SOLUTION**

The digit sum is $1 + X + 2 + X + 3 = 6 + 2X$. For divisibility by 9, $6 + 2X$ must be a multiple of 9. From the choices, $X = 6$ gives 18.

QUESTION 2

If we define $a \star b = a^2 - b^2$, what is $5 \star 4$?

A 1

B 7

C 9

D 16

E 25

F 41

WORKED SOLUTION

Apply the definition directly: $5 \star 4 = 5^2 - 4^2 = 25 - 16 = 9$.

QUESTION 3

Jack tosses two fair dice. What is the probability that the sum of the two values is no less than 9?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number of all possible cases}}.$$

A $\frac{1}{4}$

B $\frac{5}{18}$

C $\frac{1}{3}$

D $\frac{5}{12}$

E $\frac{2}{3}$

WORKED SOLUTION

For two dice, sums at least 9 are 9, 10, 11, 12, giving $4 + 3 + 2 + 1 = 10$ outcomes. So the probability is $10/36 = 5/18$.

M1-04

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 4

How many 4-digit integers $\overline{abcd}$ satisfy that

- 1 The thousands digit a and the tens digit c are even numbers;
- 2 The hundreds digit b and the unit digit d are odd numbers.
- 3 This number is divisible by 5?

A 80

B 100

C 125

D 250

E 500

WORKED SOLUTION

Since the number is divisible by 5, the final digit must be 5 because d is odd.

Then $a \in \{2, 4, 6, 8\}$, b has 5 choices, and c has 5 choices, giving

$$4 \cdot 5 \cdot 5 \cdot 1 = 100.$$

M1-05

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 5

If $\frac{m}{10} = \frac{6}{n} = \frac{2}{5}$, what is $m + n$?

A 10

B 13

C 14

D 16

E 19

WORKED SOLUTION

From $\frac{m}{10} = \frac{2}{5}$, $m = 4$. From $\frac{6}{n} = \frac{2}{5}$, $30 = 2n$, so $n = 15$. Hence $m + n = 19$.

QUESTION 6

Nancy has 100 marbles. She gives 20% of them to her friend Pedro. Then Nancy gives 25% of what is left to another friend, Ben. Finally, Nancy gives 10% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Nancy have left for herself?

A 20

B 30

C $33\frac{1}{3}$

D 38

E 45

F 50

G 54

H 60

WORKED SOLUTION

Track what remains: after 20% is given away, 80 remain; after giving 25% of 80, 60 remain; after giving 10% of 60, 54 remain.

QUESTION 7

What is the sum of all positive 2-digit integers that have a remainder of 1 when divided by 9 and a remainder of 2 when divided by 5?

A 29

B 56

C 84

D 119

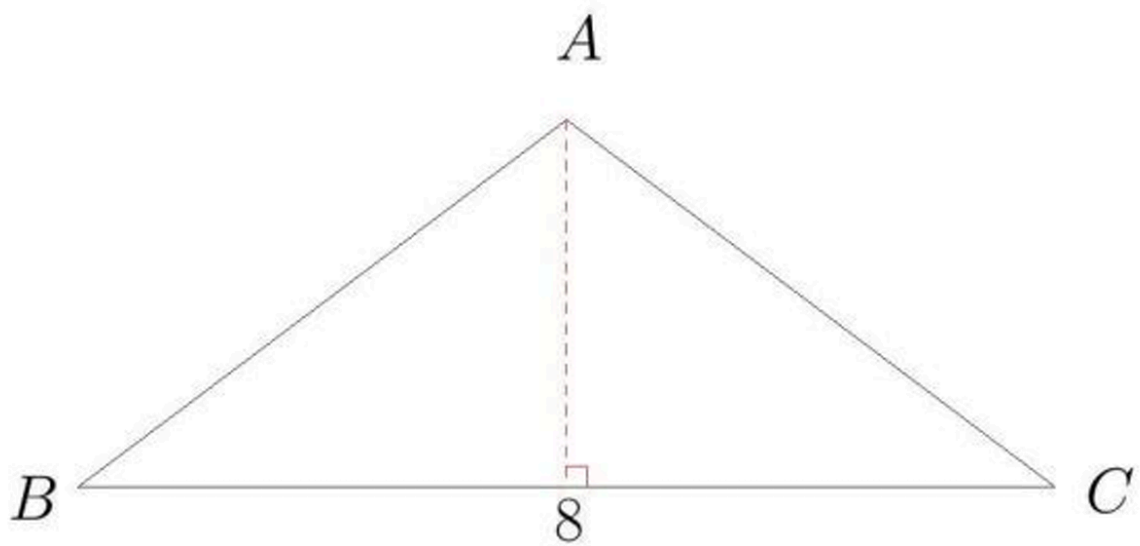
E 164

WORKED SOLUTION

The numbers satisfying both conditions are 37 and 82: each is 1 more than a multiple of 9 and 2 more than a multiple of 5. Their sum is 119.

QUESTION 8

$\triangle ABC$ is an isosceles triangle. $AB = AC$. If $BC = 8$ and the area of $\triangle ABC$ is 12, what is the perimeter of $\triangle ABC$?



A 12

B 16

C 18

D 19

E 20

WORKED SOLUTION

The height is h where $\frac{1}{2} \cdot 8 \cdot h = 12$, so $h = 3$. The altitude bisects the base, giving half-base 4, so each equal side is $\sqrt{3^2 + 4^2} = 5$. Perimeter $= 8 + 5 + 5 = 18$.

QUESTION 9

What is the smaller angle between the hour hand and minute hand of a clock at 5:50 pm?

A 100°

B 125°

C 150°

D 155°

E 165°

WORKED SOLUTION

At 5:50, the minute hand is at 300° . The hour hand is at $5 \cdot 30 + 50 \cdot 0.5 = 175^\circ$. The smaller angle is $300 - 175 = 125^\circ$.

QUESTION 10

The average age of 5 people in a room is 40 years. A 36-year-old person leaves the room. What is the average age of the four remaining people?

A 39

B 40

C 41

D 43

E 45

WORKED SOLUTION

The original total age is $5 \cdot 40 = 200$. After the 36-year-old leaves, the total is 164, so the new average is $164/4 = 41$.

M1-11

Mock Test 1 - Mathematics 1

Answer **A**

QUESTION 11

In how many ways can we write 127 as the sum of two prime numbers?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

An odd number written as a sum of two primes must use 2 plus an odd prime. Here $127 = 2 + 125$, but 125 is not prime, so there are no ways.

QUESTION 12

If for some positive integer n ,

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \cdots \left(1 + \frac{1}{n}\right) = 50,$$

what is n ?

A 20

B 39

C 65

D 99

E 100

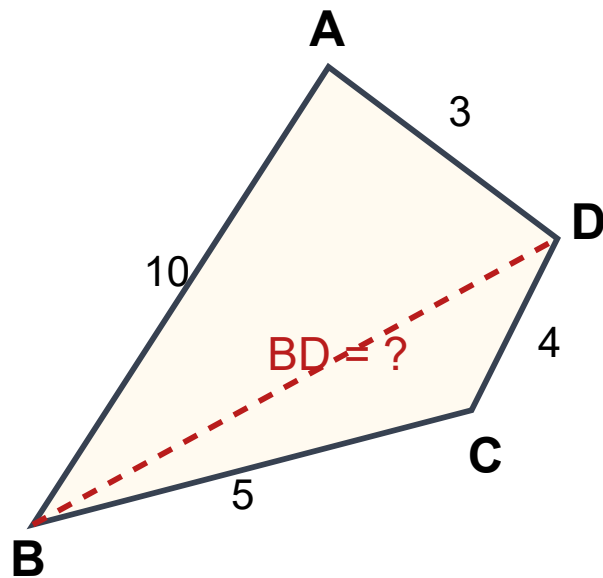
WORKED SOLUTION

Each factor is $1 + \frac{1}{k} = \frac{(k+1)}{k}$. The product telescopes:

$$\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2} = 50. \text{ Thus } n = 99.$$

QUESTION 13

In quadrilateral $ABCD$, $AB = 10$, $BC = 5$, $CD = 4$ and $AD = 3$. If the length of BD is an integer, what is BD ?



A 7

B 8

C 9

D 10

E there is not enough information

WORKED SOLUTION

Use the triangle inequality in the two triangles that share BD . For $\triangle ABD$, $|10 - 3| < BD < 10 + 3$, so $7 < BD < 13$. For $\triangle BCD$, $|5 - 4| < BD < 5 + 4$, so $1 < BD < 9$. The only integer satisfying both is $BD = 8$.

QUESTION 14

What is the unit digit of 7^{2020} ? Here

$$7^n = \underbrace{7 \times 7 \times 7 \times \dots \times 7}_n$$

(Hint: check unit digits of 7 , 7^2 , 7^3 , ... and find the pattern.)

A 1**B** 3**C** 5**D** 7**E** 9**WORKED SOLUTION**

The units digits of powers of 7 cycle as 7, 9, 3, 1. Since 2020 is divisible by 4, the units digit is the fourth term of the cycle, 1.

QUESTION 15

Here x , y , z and u represent four different numbers.

$$\begin{array}{r} x \quad y \quad x \quad y \\ + z \quad x \quad - z \quad x \\ \hline u \quad x \quad \quad x \end{array}$$

$$x + y + z + u =$$

A 12

B 14

C 16

D 18

E 20

WORKED SOLUTION

From the subtraction, $(10x + y) - (10z + x) = x$, so $8x + y = 10z$. From the addition, $(10x + y) + (10z + x) = 10u + x$, so $10x + y + 10z = 10u$. The distinct digit solution is $(x, y, z, u) = (5, 0, 4, 9)$, giving 18.

M1-16

Mock Test 1 - Mathematics 1

Answer C

QUESTION 16

Billy, a seasonal worker in the town of Cowra, collected 8 buckets of cherries on his first day. Each day after that he increased the number of buckets he picked by 2 buckets per day. According to this pattern, how many buckets will he collect in the first 20 days?

A 308

B 500

C 540

D 580

E 600

WORKED SOLUTION

The buckets form an arithmetic sequence: 8, 10, 12, The 20th term is $8 + 19 \cdot 2 = 46$. Sum = $\frac{20}{2}(8 + 46) = 540$.

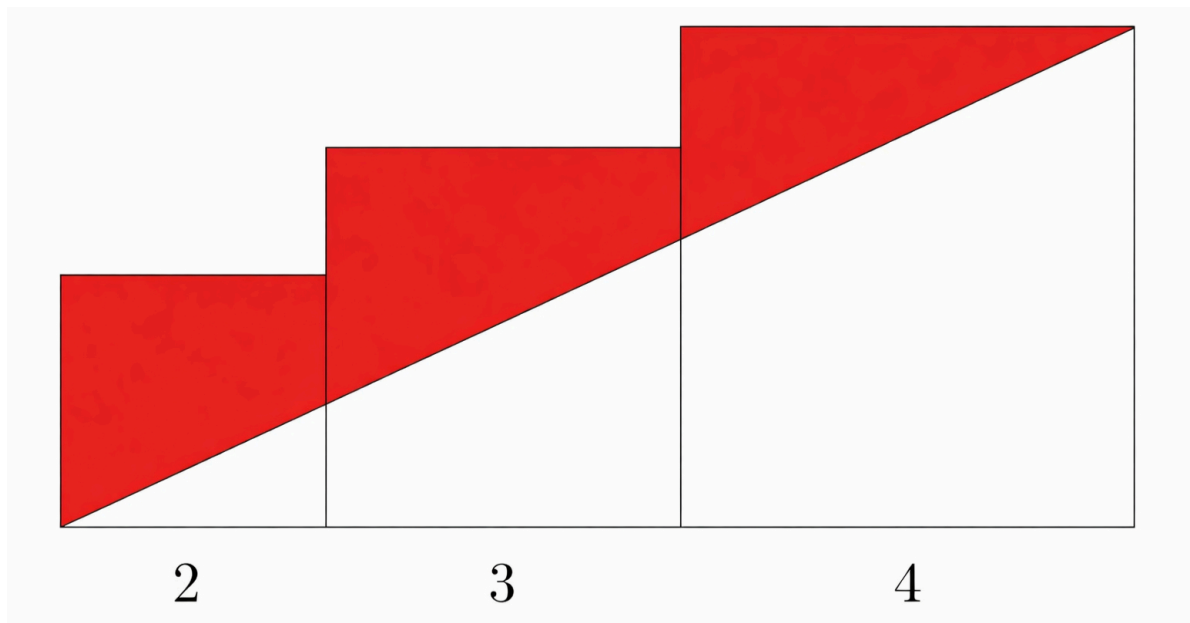
M1-17

Mock Test 1 - Mathematics 1

Answer **B**

QUESTION 17

Three squares are lined up horizontally as shown. Their side lengths are 2, 3 and 4 respectively. A straight line is drawn from the top right corner of the largest square to the bottom left hand corner of the smallest square. What is the area of the shaded region?



A 10

B 11

C 13

D 16

E 18

WORKED SOLUTION

The total area of the three squares is $2^2 + 3^2 + 4^2 = 29$. The line cuts off one large triangle of base 9 and height 4, area 18. Shaded area = $29 - 18 = 11$.

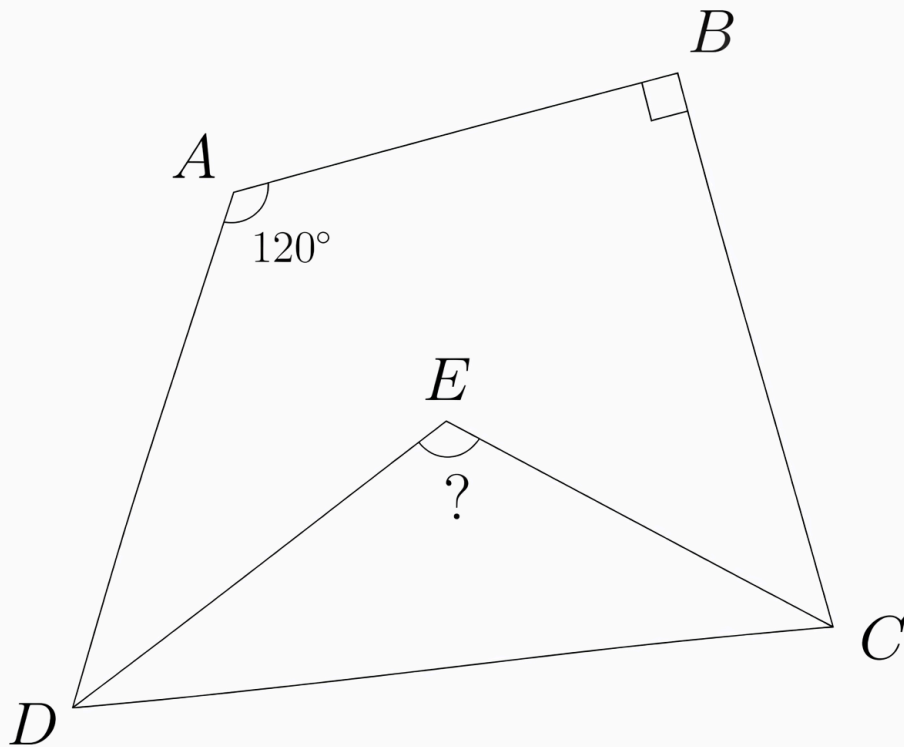
M1-18

Mock Test 1 - Mathematics 1

Answer D

QUESTION 18

In quadrilateral $ABCD$, $\angle A = 120^\circ$ and $\angle B = 90^\circ$. ED and EC bisect $\angle ADC$ and $\angle BCD$ respectively. What is $\angle E$?



A 90

B 95

C 100

D 105

E 110

WORKED SOLUTION

In the quadrilateral, $\angle C + \angle D = 360^\circ - 120^\circ - 90^\circ = 150^\circ$. Since EC and ED bisect those angles, triangle DCE has angles at C, D summing to 75° .
Therefore $\angle E = 180^\circ - 75^\circ = 105^\circ$.

M1-19

Mock Test 1 - Mathematics 1

Answer **C**

QUESTION 19

In a mathematics contest with 10 problems, a student gains 6 points for a correct answer and loses 1 point for an incorrect answer. If Olivia answered every problem and her score was 32, how many correct answers did she have?

A 4

B 5

C 6

D 7

E 8

WORKED SOLUTION

Let c be the number correct. Then incorrect answers are $10 - c$, so
 $6c - (10 - c) = 32$. Hence $7c = 42$ and $c = 6$.

M1-20

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 20

Emily and Owen are going to a library that is 1 mile from their house. They leave home simultaneously. Emily rides her bicycle to the library at a constant speed of 12 miles per hour. Owen walks to the library at a constant speed of 5 miles per hour. How many minutes before Owen does Emily arrive?

A 3

B 4

C 5

D 6

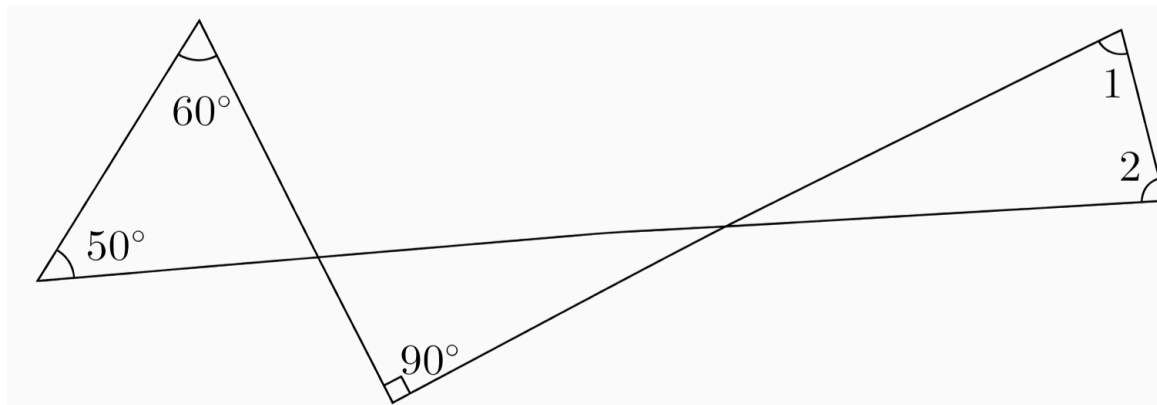
E 7

WORKED SOLUTION

Emily takes $1/12$ hour = 5 minutes. Owen takes $1/5$ hour = 12 minutes. Emily arrives $12 - 5 = 7$ minutes earlier.

QUESTION 21

In the following picture, if $\angle 1 = \angle 2$, what is $\angle 1$?



A 60°

B 65°

C 70°

D 75°

E 80°

WORKED SOLUTION

The left triangle gives the angle at the first intersection as

$180^\circ - 50^\circ - 60^\circ = 70^\circ$. The next triangle has a right angle, so the angle at the second intersection is 20° . The final small triangle has angles x , x , 20° , so $2x = 160^\circ$, giving $x = 80^\circ$.

QUESTION 22

Jack, a zoo keeper, is going to give N peaches to a group of monkeys. If he gives 10 peaches to each monkey, two monkeys will receive no peach. Or, the peaches are exactly enough to give 8 to each monkey. What is the sum of digits of N ?

A 5

B 6

C 8

D 9

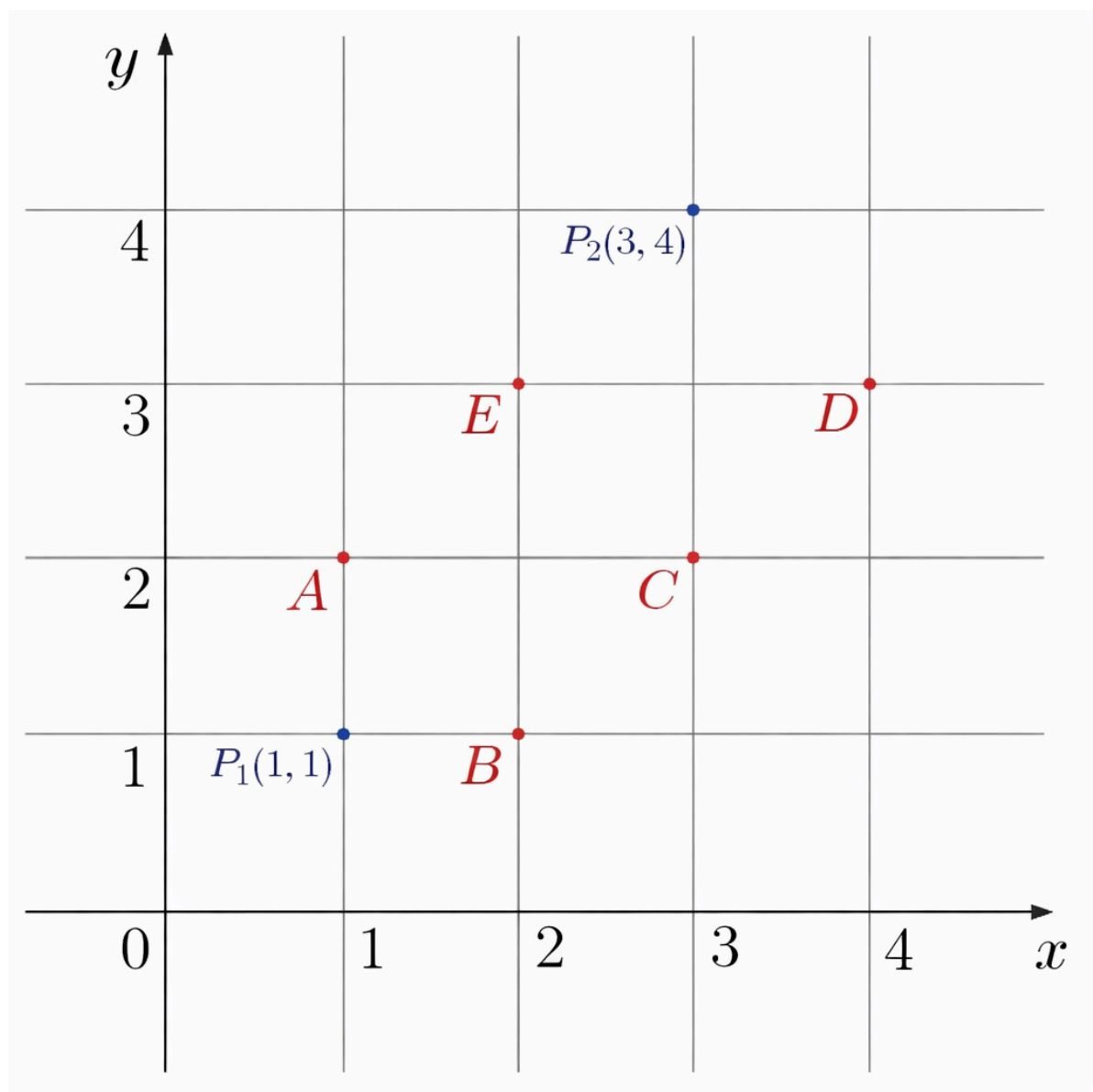
E 10

WORKED SOLUTION

Let there be m monkeys. The two descriptions give $N = 8m$ and $N = 10(m - 2)$. Thus $8m = 10m - 20$, so $m = 10$ and $N = 80$. Digit sum = 8.

QUESTION 23

Every point on the plane can be assigned with a coordinate that has two numbers. For example, the coordinate of point P_1 in the following picture is $(1, 1)$ and the coordinate of point P_2 is $(3, 4)$. Which of the following points has the coordinate $(2, 3)$?



A A

B B

C C

D D

E E

WORKED SOLUTION

Read the grid directly: the point at (2, 3) is labelled E.

M1-24

Mock Test 1 - Mathematics 1

Answer **D**

QUESTION 24

A pizza-shop offers a basic version of pizza with mozzarella and tomatoes. Two different toppings from the following list must be added: anchovies, artichokes, mushrooms, capers. Moreover, for each pizza three different sizes are available: small, medium, large. How many different types of pizza are available at all?

A 11

B 12

C 15

D 18

E 36

WORKED SOLUTION

Choose 2 toppings from 4: $\binom{4}{2} = 6$. For each, there are 3 sizes, so total pizzas $= 6 \cdot 3 = 18$.

M1-25

Mock Test 1 - Mathematics 1

Answer **E**

QUESTION 25

Bag A: 1 Snickers bar + 1 Butterfinger bar.

Bag B: 1 Butterfinger bar + 3 Reese's Peanut Butter Cups.

Bag C: 1 Snickers bar + 2 Reese's Peanut Butter Cups.

If each bag costs 50 cents, what is the price of
1 Snickers bar + 1 Butterfinger bar + 1 Reese's Peanut Butter Cup,
in cents?

A 40

B 45

C 50

D 55

E 60

WORKED SOLUTION

Let the prices be S , B , R . The bags give $S + B = 50$, $B + 3R = 50$, and $S + 2R = 50$. Solving gives $S = 30$, $B = 20$, $R = 10$, so $S + B + R = 60$.

M1-26

Mock Test 1 - Mathematics 1

Answer C

QUESTION 26

For how many positive integers x is

$$(x - 1)(x - 4)(x - 9)$$

negative?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

The roots are 1, 4 and 9. Testing the sign in the intervals shows the product is negative for $x < 1$ and $4 < x < 9$. The positive integers satisfying this are 5, 6, 7, 8, so there are 4 values.

M1-27

Mock Test 1 - Mathematics 1

Answer **C**

QUESTION 27

A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is

A 8

B 9

C 10

D 12

E 18

WORKED SOLUTION

The most gumballs possible without having four of one colour is $3 + 3 + 3 = 9$.
One more guarantees four of some colour, so the least number is 10.

Mathematics 2

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01

Mock Test 1 - Mathematics 2

Answer A

QUESTION 1

Rearrange

$$\frac{7x + 10}{9x + 5} = 3y^2 + 2$$

to make x the subject.

A

$$\frac{15y^2}{7 - 9(3y^2 + 2)}$$

B

$$\frac{15y^2}{7 + 9(3y^2 + 2)}$$

C

$$- \frac{15y^2}{7 - 9(3y^2 + 2)}$$

D

$$- \frac{15y^2}{7 + 9(3y^2 + 2)}$$

E

$$- \frac{5y^2}{7 + 9(3y^2 + 2)}$$

F

$$\frac{5y^2}{7 + 9(3y^2 + 2)}$$

WORKED SOLUTION

Let $A = 3y^2 + 2$. Then $7x + 10 = A(9x + 5)$, so
 $x(7 - 9A) = 5A - 10 = 15y^2$. Therefore $x = \frac{15y^2}{7 - 9(3y^2 + 2)}$.

M2-02

Mock Test 1 - Mathematics 2

Answer **F**

QUESTION 2

Simplify

$$3x \left(\frac{3x^7}{\frac{1}{x^3}} \right)^3$$

A $9x^{20}$

B $27x^{20}$

C $87x^{20}$

D $9x^{21}$

E $27x^{21}$

F $81x^{31}$

WORKED SOLUTION

Dividing by $1/x^3$ is the same as multiplying by x^3 , so the bracket becomes $3x^{10}$.
Cubing gives $27x^{30}$, and multiplying by $3x$ gives $81x^{31}$.

QUESTION 3

For $x \geq 0$, simplify

$$2x[(2x)^7]^{\frac{1}{14}}$$

A $2x\sqrt{2x^4}$

B $2x\sqrt{2x^3}$

C $2\sqrt{2x^4}$

D $2\sqrt{2x^3}$

E $8x^3$

F $8x$

WORKED SOLUTION

$[(2x)^7]^{1/14} = (2x)^{1/2} = \sqrt{2x}$. Then $2x\sqrt{2x} = 2\sqrt{2x^3}$ for $x \geq 0$.

QUESTION 4

What is the circumference of a circle with an area of 10π ?

A $2\pi\sqrt{10}$

B $\pi\sqrt{10}$

C 10π

D 20π

E $\sqrt{10}$

F More information needed.

WORKED SOLUTION

Area = $\pi r^2 = 10\pi$, so $r = \sqrt{10}$. Circumference = $2\pi r = 2\pi\sqrt{10}$.

M2-05

Mock Test 1 - Mathematics 2

Answer **D**

QUESTION 5

If $a \cdot b = (ab) + (a + b)$, then calculate the value of $(3 \cdot 4) \cdot 5$.

A 19

B 54

C 100

D 119

E 132

WORKED SOLUTION

First, $3 \cdot 4 = 3 \cdot 4 + (3 + 4) = 12 + 7 = 19$. Then
 $19 \cdot 5 = 19 \cdot 5 + (19 + 5) = 95 + 24 = 119$.

M2-06

Mock Test 1 - Mathematics 2

Answer **D**

QUESTION 6

If $a \cdot b = \frac{a^b}{a}$, calculate $(2 \cdot 3) \cdot 2$.

A $\frac{16}{3}$

B 1

C 2

D 4

E 8

WORKED SOLUTION

$2 \cdot 3 = 2^3 / 2 = 4$. Then $4 \cdot 2 = 4^2 / 4 = 4$.

M2-07

Mock Test 1 - Mathematics 2

Answer **F**

QUESTION 7

Solve $x^2 + 3x - 5 = 0$.

A $x = -\frac{3}{2} \pm \frac{\sqrt{11}}{2}$

B $x = \frac{3}{2} \pm \frac{\sqrt{11}}{2}$

C $x = -\frac{3}{2} \pm \frac{\sqrt{11}}{4}$

D $x = \frac{3}{2} \pm \frac{\sqrt{11}}{4}$

E $x = \frac{3}{2} \pm \frac{\sqrt{29}}{2}$

F $x = -\frac{3}{2} \pm \frac{\sqrt{29}}{2}$

WORKED SOLUTION

Using the quadratic formula, $x = \frac{-3 \pm \sqrt{3^2 - 4(1)(-5)}}{2} = \frac{-3 \pm \sqrt{29}}{2}$.

M2-08

Mock Test 1 - Mathematics 2

Answer **B**

QUESTION 8

How many times do the curves $y = x^3$ and $y = x^2 + 4x + 14$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Intersections satisfy $x^3 = x^2 + 4x + 14$, or $x^3 - x^2 - 4x - 14 = 0$. This cubic has one real root and two complex roots, so the curves intersect once.

M2-09

Mock Test 1 - Mathematics 2

Answer D**QUESTION 9**

Solve for x , y , and z .

1. $x + y - z = -1$
2. $2x - 2y + 3z = 8$
3. $2x - y + 2z = 9$

	x	y	z
A	2	-15	-14
B	15	2	14
C	14	15	-2
D	-2	15	14
E	2	-15	14
F	No solutions possible		

WORKED SOLUTION

Substitute option D: $x = -2$, $y = 15$, $z = 14$. Then $-2 + 15 - 14 = -1$, $-4 - 30 + 42 = 8$, and $-4 - 15 + 28 = 9$. It satisfies all three equations.

M2-10

Mock Test 1 - Mathematics 2

Answer D

QUESTION 10Fully factorise: $3a^3 - 30a^2 + 75a$

A $3a(a - 3)^3$

B $a(3a - 5)^2$

C $3a(a^2 - 10a + 25)$

D $3a(a - 5)^2$

E $3a(a + 5)^2$

WORKED SOLUTIONFactor out $3a$: $3a^3 - 30a^2 + 75a = 3a(a^2 - 10a + 25) = 3a(a - 5)^2$.**M2-11**

Mock Test 1 - Mathematics 2

Answer B

QUESTION 11Solve for x and y :

$4x + 3y = 48$

$3x + 2y = 34$

	x	y
A	8	6
B	6	8

	x	y
C	3	4
D	4	3
E	30	12
F	12	30
G	No solutions possible	

WORKED SOLUTION

Eliminate y : double the first equation gives $8x + 6y = 96$; triple the second gives $9x + 6y = 102$. Subtract to get $x = 6$, then $3(6) + 2y = 34$ gives $y = 8$.

M2-12

Mock Test 1 - Mathematics 2

Answer E

QUESTION 12

Evaluate:

$$- \frac{(5^2 - 4 \times 7)^2}{-6^2 + 2 \times 7}$$

A $-\frac{3}{50}$

B $\frac{11}{22}$

C $-\frac{3}{22}$

D $\frac{9}{50}$

E $\frac{9}{22}$

F 0

WORKED SOLUTION

The numerator is $-(25 - 28)^2 = -9$. The denominator is $-6^2 + 14 = -36 + 14 = -22$. Hence the value is $(-9)/(-22) = 9/22$.

M2-13

Mock Test 1 - Mathematics 2

Answer E

QUESTION 13

All license plates are 6 characters long. The first 3 characters consist of letters and the next 3 characters of numbers. Letters and digits may be repeated. How many unique license plates are possible?

A 676,000

B 6,760,000

C 67,600,000

D 1,757,600

E 17,576,000

F 175,760,000

WORKED SOLUTION

There are 26^3 ways to choose the letters and 10^3 ways to choose the digits. Total $26^3 \cdot 10^3 = 17,576,000$.

M2-14

Mock Test 1 - Mathematics 2

Answer **B**

QUESTION 14

How many solutions are there for:

$$2(2(x^2 - 3x)) = -9?$$

A 0

B 1

C 2

D 3

E Infinite solutions.

WORKED SOLUTION

The equation becomes $4x^2 - 12x = -9$, or $4x^2 - 12x + 9 = 0$. The discriminant is $(-12)^2 - 4(4)(9) = 0$, so there is exactly one solution.

M2-15

Mock Test 1 - Mathematics 2

Answer **C**

QUESTION 15

Evaluate:

$$(x^{\frac{1}{2}}y^{-3})^{\frac{1}{2}}$$

A $\frac{\frac{1}{x^2}}{y}$

B $\frac{x}{\frac{3}{y^2}}$

C $\frac{\frac{1}{x^4}}{\frac{3}{y^2}}$

D $\frac{\frac{1}{y^4}}{\frac{3}{x^2}}$

WORKED SOLUTION

Use index laws: $(x^{1/2}y^{-3})^{1/2} = x^{1/4}y^{-3/2} = \frac{x^{1/4}}{y^{3/2}}$.

M2-16

Mock Test 1 - Mathematics 2

Answer A

QUESTION 16

Bryan earned a total of £ 1,240 last week from renting out three flats. From this, he had to pay 10% of the rent from the 1-bedroom flat for repairs, 20% of the rent from the 2-bedroom flat for repairs, and 30% from the 3-bedroom flat for repairs.

The 3-bedroom flat costs twice as much as the 1-bedroom flat. Given that the total repair bill was £ 276, calculate the rent for each apartment.

	1 Bedroom	2 Bedrooms	3 Bedrooms
A	280	400	560
B	140	200	280
C	420	600	840
D	250	300	500

WORKED SOLUTION

Let the 1-bedroom rent be a , the 2-bedroom rent be b , and the 3-bedroom rent be $2a$. Then $3a + b = 1240$, and repairs give $0.1a + 0.2b + 0.3(2a) = 276$.

Solving gives $a = 280$, $b = 400$, and $2a = 560$.

M2-17

Mock Test 1 - Mathematics 2

Answer D

QUESTION 17

At a Pizza Parlour, you can order single, double or triple cheese in the crust. You also have the option to include ham, olives, pepperoni, bell pepper, meat balls, tomato slices, and pineapples. How many different types of pizza are available at the Pizza Parlour?

A 10**B** 96**C** 192**D** 384**E** 768**F** None of the above**WORKED SOLUTION**

There are 3 cheese choices. Each of the 7 toppings can be included or not included, giving 2^7 topping combinations. Total = $3 \cdot 2^7 = 384$.

QUESTION 18

Solve the simultaneous equations $x^2 + y^2 = 1$ and $x + y = \sqrt{2}$, for $x, y > 0$.

A $(x, y) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$

B $(x, y) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$

C $(x, y) = (\sqrt{2} - 1, 1)$

D $(x, y) = \left(\sqrt{2}, \frac{1}{2} \right)$

WORKED SOLUTION

Square $x + y = \sqrt{2}$: $x^2 + y^2 + 2xy = 2$. Since $x^2 + y^2 = 1$, $2xy = 1$. With positive x, y and sum $\sqrt{2}$, this gives $x = y = \sqrt{2}/2$.

QUESTION 19

Which of the following statements is **FALSE**?

A Congruent objects always have the same dimensions and shape.

B Congruent objects can be mirror images of each other.

C Congruent objects do not always have the same angles.

D Congruent objects can be rotations of each other.

E Two triangles are congruent if they have two sides and the included angle of the same magnitude.

WORKED SOLUTION

Congruent shapes have the same side lengths and the same angles. They may be translated, rotated or reflected, but not enlarged or shrunk. Therefore statement C is false.

M2-20

Mock Test 1 - Mathematics 2

Answer **B**

QUESTION 20

Solve the inequality $x^2 \geq 6 - x$.

A $x \leq -3$ and $x \leq 2$

B $x \leq -3$ and $x \geq 2$

C $x \geq -3$ and $x \leq 2$

D $x \geq -3$ and $x \geq 2$

E $x \geq 2$ only

F $x \geq -3$ only

WORKED SOLUTION

Move all terms to one side: $x^2 + x - 6 \geq 0$. Factor: $(x + 3)(x - 2) \geq 0$. The upward-opening quadratic is non-negative outside the roots, so $x \leq -3$ or

$$x \geq 2.$$

M2-21

Mock Test 1 - Mathematics 2

Answer A

QUESTION 21

The hypotenuse of an isosceles right-angled triangle is x cm. What is the area of the triangle in terms of x ?

A $\frac{x^2}{4}$

B $\frac{x}{4}$

C $\frac{3x^2}{4}$

D $\frac{x^2}{10}$

WORKED SOLUTION

Let each equal leg be a . Then $a^2 + a^2 = x^2$, so $a^2 = x^2/2$. Area = $\frac{1}{2}a^2 = \frac{x^2}{4}$.

M2-22

Mock Test 1 - Mathematics 2

Answer A

QUESTION 22

Mr Heard derives a formula:

$$Q = \frac{(X + Y)^2 A}{3B}$$

He doubles the values of X and Y , halves the value of A and triples the value of B .
What happens to the value of Q ?

A Decreases by $\frac{1}{3}$

B Increases by $\frac{1}{3}$

C Decreases by $\frac{2}{3}$

D Increases by $\frac{2}{3}$

E Increases by $\frac{4}{3}$

F Decreases by $\frac{4}{3}$

WORKED SOLUTION

Doubling both X and Y makes $(X + Y)^2$ four times larger. Halving A makes the factor 2, and tripling B divides by 3. New $Q = (2/3)$ of old Q , so it decreases by $1/3$.

M2-23

Mock Test 1 - Mathematics 2

Answer **C**

QUESTION 23

Consider the graphs $y = x^2 - 2x + 3$, and $y = x^2 - 6x - 10$. Which of the following is true?

A Both equations intersect the x -axis.

B Neither equation intersects the x -axis.

C The first equation does not intersect the x -axis; the second equation intersects the x -axis.

- D** The first equation intersects the x-axis; the second equation does not intersect the x-axis.

WORKED SOLUTION

Use the discriminant. For $x^2 - 2x + 3$, $b^2 - 4ac = 4 - 12 = -8 < 0$, so no x-intercepts. For $x^2 - 6x - 10$, $36 + 40 = 76 > 0$, so it intersects the x-axis.

M2-24

Mock Test 1 - Mathematics 2

Answer C

QUESTION 24

Evaluate

$$\log_2\left(\frac{5}{4}\right) + \log_2\left(\frac{6}{5}\right) + \log_2\left(\frac{7}{6}\right) + \dots + \log_2\left(\frac{64}{63}\right)$$

A -2

B 3

C 4

D 6

WORKED SOLUTION

The logarithms telescope:

$$\log_2(5/4) + \log_2(6/5) + \dots + \log_2(64/63) = \log_2(64/4) = \log_2(16) = 4.$$

QUESTION 25

Given that $7 \cos \theta - 3 \tan \theta \sin \theta = 1$, which one of the following is true?

A $\cos \theta = -\frac{3}{5}$ or $\cos \theta = -\frac{1}{2}$

B $\cos \theta = -\frac{3}{5}$ or $\cos \theta = \frac{1}{2}$

C $\cos \theta = \frac{3}{5}$ or $\cos \theta = \frac{1}{2}$

D $\cos \theta = \frac{3}{5}$ or $\cos \theta = -\frac{1}{2}$

WORKED SOLUTION

Let $c = \cos \theta$. Since $\tan \theta \sin \theta = \frac{\sin^2 \theta}{\cos \theta} = \frac{(1 - c^2)}{c}$, multiply the equation by c : $7c^2 - 3(1 - c^2) = c$. Thus

$$10c^2 - c - 3 = 0 = (5c - 3)(2c + 1), \text{ so } c = \frac{3}{5} \text{ or } c = -\frac{1}{2}.$$

QUESTION 26

k is the smallest positive value of x which is a solution to both the equations $2 \sin x + 1 = 0$ and $2 \cos 2x = 1$.

How many values of x in the range $0 \leq x \leq k$ are solutions to at least one of these equations?

A 0

B 2

C 3

D 4

E 8

WORKED SOLUTION

$2 \sin x + 1 = 0$ gives $\sin x = -1/2$. The first positive common solution with $2 \cos 2x = 1$ is $k = 7\pi/6$. In $[0, k]$, the solutions to at least one equation are $\pi/6$, $5\pi/6$, and $7\pi/6$: three values.

M2-27

Mock Test 1 - Mathematics 2

Answer D

QUESTION 27

How many solutions of the equation $2 \sin^3 \theta = \sin \theta$ lie in the interval $-\frac{\pi}{2} \leq \theta \leq \pi$?

A 2

B 3

C 4

D 5

E 6

F 7

WORKED SOLUTION

Factor: $2 \sin^3 \theta = \sin \theta \Rightarrow \sin \theta (2 \sin^2 \theta - 1) = 0$. In the interval, $\sin \theta = 0$ gives $0, \pi$; $\sin \theta = \sqrt{2}/2$ gives $\pi/4, 3\pi/4$; $\sin \theta = -\sqrt{2}/2$ gives $-\pi/4$. Total 5 solutions.

Biology

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Biology Easy source set. Question wording, option order, answer and solution method are unchanged.

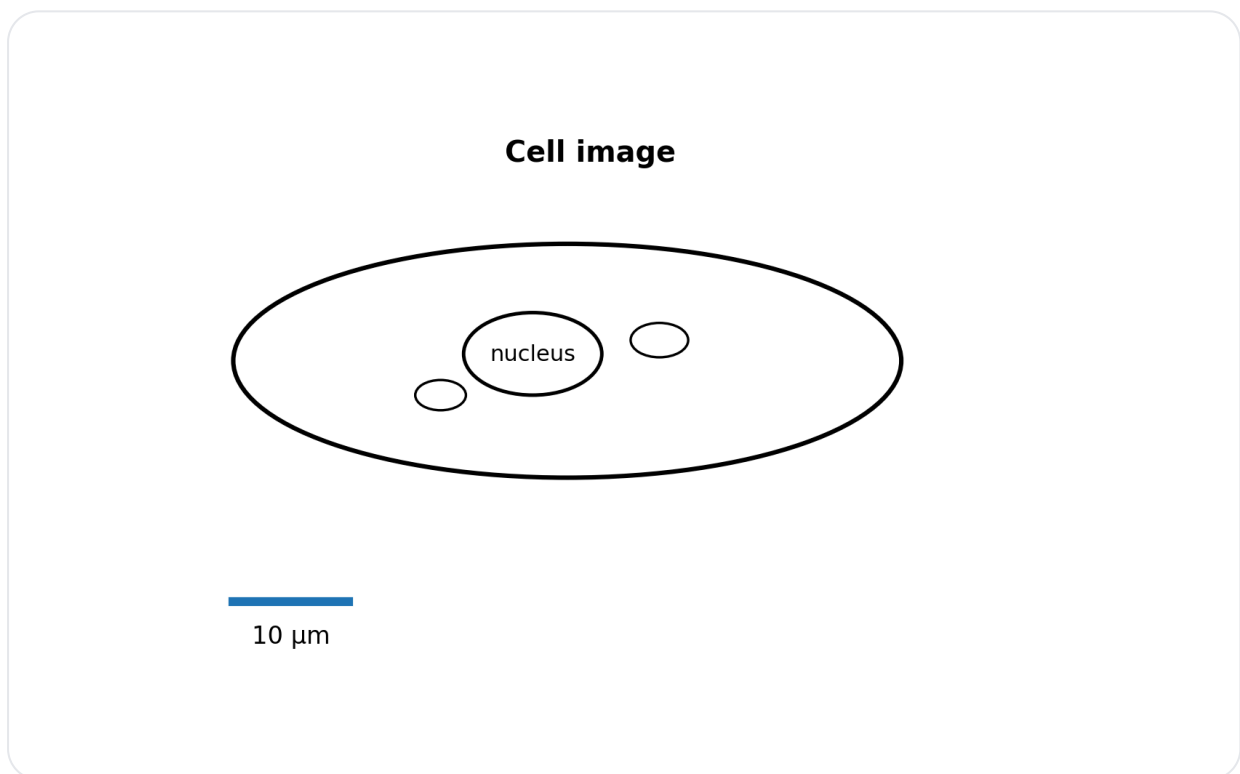
B-01

Microscopy

Answer C

QUESTION 1

The diagram shows a cell viewed using a microscope. The scale bar represents **10 μm** .



Approximately how long is the cell along its longest axis?

A 20 μm

B 40 μm

C 60 μm

D 80 μm

E 100 μm

WORKED SOLUTION

The cell is approximately 5.4 scale-bar lengths long.

$$5.4 \times 10 \mu\text{m} \approx 54 \mu\text{m}$$

The nearest option is **60 μm** .

B-02

Levels of organisation

Answer **A**

QUESTION 2

Which sequence places the structures in order from **smallest to largest**?

1. circulatory system
2. heart
3. mitochondrion
4. muscle cell
5. muscle tissue

A 3, 4, 5, 2, 1

B 4, 3, 5, 2, 1

C 3, 5, 4, 2, 1

D 3, 4, 2, 5, 1

E 4, 5, 3, 2, 1

WORKED SOLUTION

The hierarchy is:

sub-cellular component < cell < tissue < organ < organ system

Therefore the correct order is **mitochondrion** → **muscle cell** → **muscle tissue** → **heart** → **circulatory system**.

B-03

Gas exchange

Answer **H**

QUESTION 3

Which features increase the rate at which oxygen moves from an alveolus into the blood?

1. A large alveolar surface area.
2. A very thin barrier between alveolar air and blood.
3. Continuous blood flow carrying oxygenated blood away from the alveolus.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All three increase diffusion rate. A large surface area provides more area for exchange, a thin barrier shortens diffusion distance, and blood flow removes oxygen that has entered the blood, helping maintain a steep concentration gradient.

B-04

Active transport

Answer C

QUESTION 4

Glucose is being absorbed from the small intestine.

[glucose in intestinal lumen] < [glucose inside an epithelial cell]

but glucose continues to enter the epithelial cell.

Which row best describes this movement and predicts the effect of strongly inhibiting respiration in the epithelial cell?

A diffusion; uptake increases

B osmosis; uptake decreases

C active transport; uptake decreases

D active transport; no effect

E diffusion; no effect

WORKED SOLUTION

Glucose is moving from lower concentration outside the cell to higher concentration inside the cell, so it is moving against its concentration gradient. This requires **active transport**.

Active transport requires energy from respiration, so strongly inhibiting respiration should **decrease uptake**.

B-05

Mitosis and meiosis

Answer E

QUESTION 5

A diploid cell contains **8 chromosomes**.

Which row correctly describes the products of mitosis and meiosis?

- A** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 8 chromosomes each
- B** mitosis: four cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- C** mitosis: two cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- D** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 4 chromosomes each
- E** mitosis: two cells, 8 chromosomes each; meiosis: four cells, 4 chromosomes each

WORKED SOLUTION

Mitosis produces two daughter cells that retain the diploid chromosome number, so each has **8 chromosomes**.

Meiosis halves the chromosome number and produces four cells, so each has **4 chromosomes**.

B-06

Sex determination

Answer E

QUESTION 6

Assume that the probability of a child being genetically male or female is equal and that the sex of one child does not affect the next.

What is the probability that, in a family's **next three children**, at least one is female?

A 1/8**B** 1/4**C** 1/2**D** 3/4**E** 7/8**WORKED SOLUTION**

It is easiest to calculate the opposite event: all three children are male.

$$P(\text{all male}) = (1/2)^3 = 1/8$$

$$P(\text{at least one female}) = 1 - 1/8 = 7/8$$

B-07

Inheritance

Answer C

QUESTION 7

In a plant species, tall stems are controlled by a dominant allele **T**, while short stems are controlled by the recessive allele **t**.

Two tall plants produce a short offspring.

What is the probability that their next offspring will also be short?

A 0

B 1/8

C 1/4

D 1/2

E 3/4

WORKED SOLUTION

A short plant must have genotype **tt**. Since two tall plants produced a short offspring, both tall parents must be heterozygous:

$$Tt \times Tt$$

The possible genotypes are TT, Tt, Tt and tt, so:

$$P(tt) = 1/4$$

B-08

DNA base composition

Answer D

QUESTION 8

A double-stranded DNA molecule contains **200 base pairs**.

Cytosine accounts for **22%** of all bases.

How many thymine bases are present?

A 44

B 56

C 88

D 112

E 156

WORKED SOLUTION

There are 400 bases in total. In double-stranded DNA, C and G occur in equal proportions, so:

$$C = 22\%, \quad G = 22\%$$

Thus $C + G = 44\%$, leaving 56% for $A + T$. Therefore $T = 28\%$.

$$0.28 \times 400 = \mathbf{112}$$

B-09

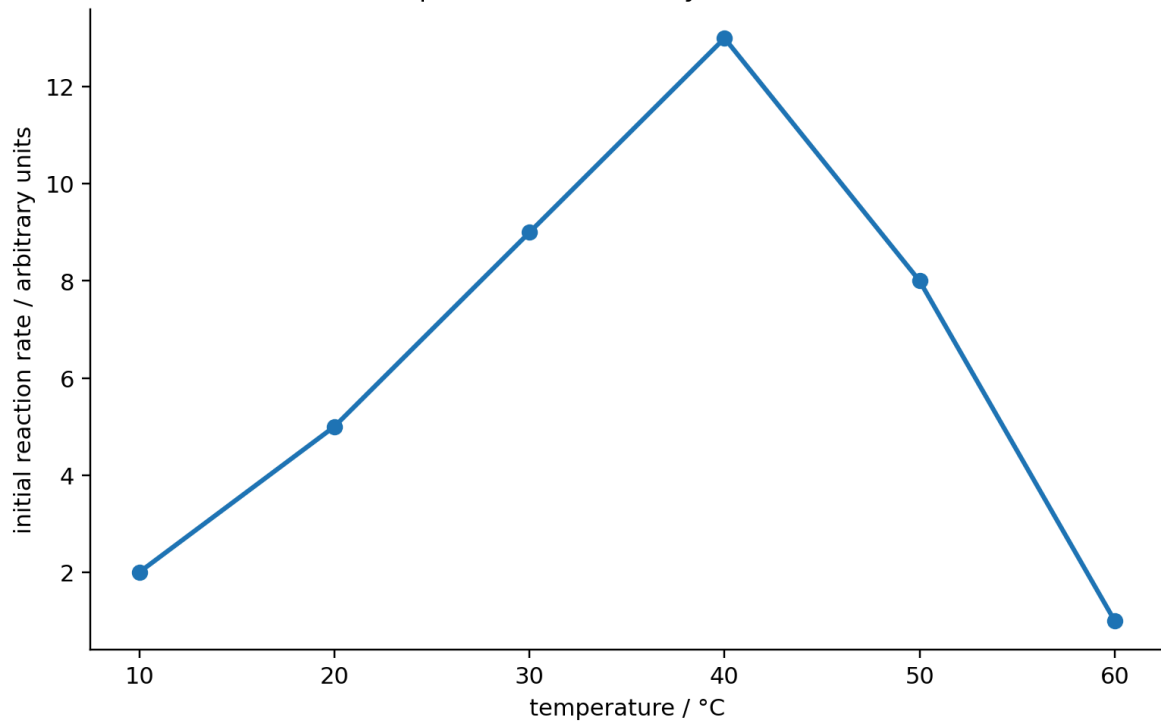
Enzymes

Answer **B**

QUESTION 9

The graph shows the initial rate of an enzyme-controlled reaction at six temperatures.

Effect of temperature on an enzyme-controlled reaction



Which statements are justified by these data?

1. The measured reaction rate increases between 10°C and 40°C.
2. The exact optimum temperature of the enzyme must be 40°C.
3. At 60°C, every enzyme molecule must have been permanently denatured.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is directly supported: the measured rate rises between 10°C and 40°C.

Statement 2 is too strong because temperatures between the measured values were not tested; the true optimum could be slightly below or above 40°C.

Statement 3 is also too strong. The graph still shows a measurable reaction at 60°C, and in any case it does not establish that every enzyme molecule has been permanently denatured.

Therefore **1 only**.

B-10

Digestion

Answer B

QUESTION 10

Which row correctly identifies an enzyme, its substrate and one of its products?

A amylase; protein; amino acids

B protease; protein; amino acids

C lipase; starch; simple sugars

D protease; lipid; fatty acids

E lipase; protein; glycerol

WORKED SOLUTION

Proteases catalyse the breakdown of proteins into amino acids. Therefore the correct combination is **protease; protein; amino acids**.

B-11

Contraception

Answer **E**

QUESTION 11

Which statements are correct?

1. Hormonal contraception containing oestrogen and progesterone can inhibit FSH production and help prevent ovulation.
2. Condoms are a non-hormonal form of contraception and can reduce transmission of sexually transmitted infections.
3. Hormonal contraceptives work mainly by directly killing sperm cells.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statements 1 and 2 are correct. Hormonal contraception can suppress FSH and therefore help prevent ovulation. Condoms are a barrier method and can also reduce transmission of many sexually transmitted infections.

Hormonal contraception does not principally work by directly killing sperm cells.

B-12

Ciliated cells

Answer **G**

QUESTION 12

Long-term smoking can damage the cilia lining parts of the respiratory tract.

Which consequences could result directly from reduced ciliary activity?

1. The percentage of oxygen in inhaled air decreases.
2. Mucus may accumulate in the airways.
3. Microorganisms trapped in mucus may be removed less effectively.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Damaged cilia do not change the percentage of oxygen in the external air being inhaled. However, mucus is cleared less effectively, so it can accumulate and microorganisms trapped in it may remain in the respiratory tract for longer.

Therefore **2 and 3 only**.

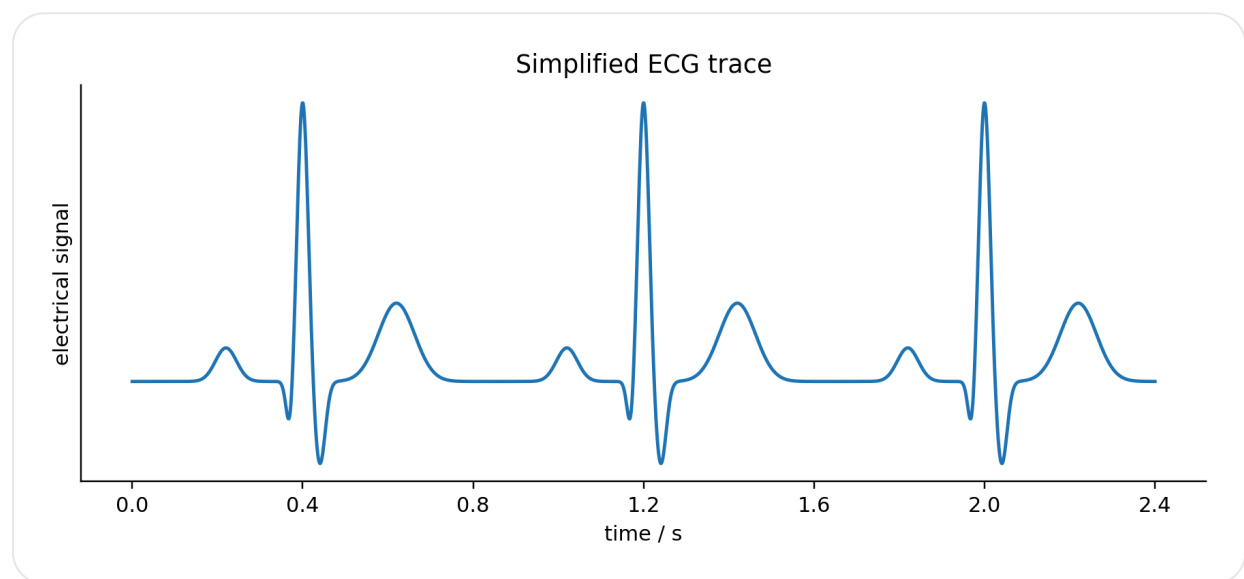
B-13

ECG and heart rate

Answer **C**

QUESTION 13

The trace shows a simplified ECG.



The large R-peaks are approximately **0.80 s** apart.

What is the heart rate?

A 48 beats min^{-1}

B 60 beats min^{-1}

C 75 beats min^{-1}

D 80 beats min^{-1}

E 96 beats min^{-1}

WORKED SOLUTION

One heartbeat takes approximately 0.80 s.

$$\text{heart rate} = 60 / 0.80 = 75 \text{ beats min}^{-1}$$

B-14

Blood

Answer D

QUESTION 14

A patient has:

- a much lower than normal platelet count;
- a much lower than normal red blood cell count.

Which combination of effects is most likely?

A blood clotting faster; oxygen-carrying capacity increased

B blood clotting faster; oxygen-carrying capacity decreased

C blood clotting slower; oxygen-carrying capacity unchanged

D blood clotting slower; oxygen-carrying capacity decreased

E blood clotting unchanged; oxygen-carrying capacity decreased

WORKED SOLUTION

Platelets are required for normal blood clotting, so a low platelet count tends to make clotting slower or less effective.

Red blood cells contain haemoglobin and carry oxygen, so a reduced red blood cell count lowers the blood's oxygen-carrying capacity.

B-15

Respiration

Answer B

QUESTION 15

Four flasks are prepared.

flask	contents	temperature
A	living yeast + glucose solution	20°C
B	living yeast + glucose solution	35°C
C	boiled yeast + glucose solution	35°C
D	living yeast + water only	35°C

Assume 35°C is closer to the optimum temperature for the enzymes involved in yeast respiration than 20°C, and neither temperature causes denaturation.

Which flask is expected to produce the greatest volume of carbon dioxide during the first ten minutes?

A A

B B

C C

D D

E A and B must produce exactly the same amount

WORKED SOLUTION

Flask C contains boiled yeast, so its cells and enzymes will no longer function normally. Flask D lacks glucose.

Both A and B contain living yeast and glucose, but B is at the stated suitable higher temperature of 35°C, so respiration should proceed faster. Therefore **B** should produce the most carbon dioxide in the first ten minutes.

B-16

Thermoregulation

Answer **B**

QUESTION 16

A healthy person's core body temperature rises above its normal value.

Which response is expected?

A sweat production decreases; skin arterioles constrict

B sweat production increases; skin arterioles dilate

C sweat production increases; skin arterioles constrict

D sweat production decreases; skin arterioles dilate

E sweat production unchanged; skin arterioles constrict

WORKED SOLUTION

When body temperature is too high, sweat production increases and skin arterioles dilate. Both changes increase heat loss from the body.

B-17

Synapses

Answer **E**

QUESTION 17

Which statements about a synapse are correct?

1. Neurotransmitters are chemicals.
2. Neurotransmitter molecules can diffuse across the synaptic gap.
3. The electrical impulse itself passes directly across the gap as an electrical current.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Neurotransmitters are chemical messengers. They are released from one neurone and diffuse across the synaptic gap.

The electrical impulse does not simply travel directly across the gap as an electrical current. Therefore **1 and 2 only**.

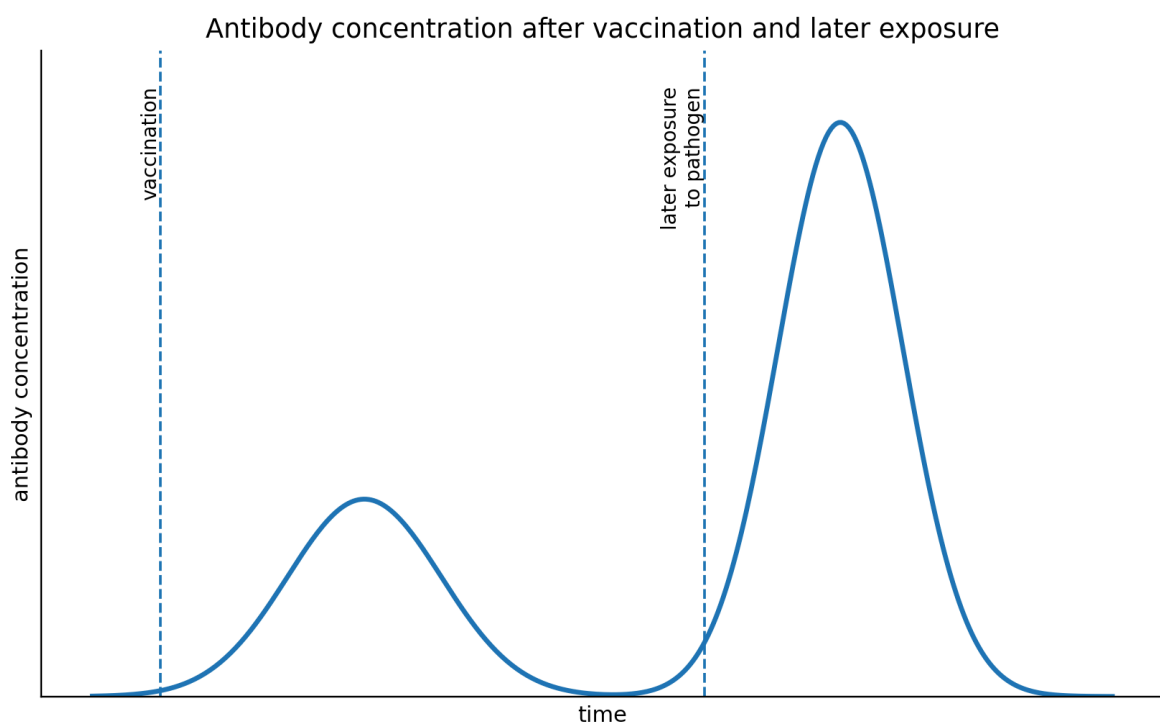
B-18

Vaccination

Answer E

QUESTION 18

The graph shows antibody concentration following vaccination and a later exposure to the same pathogen.



Which statements could explain the graph?

1. Vaccination can lead to the formation of memory cells.
2. Memory cells can allow a faster and larger antibody response during later exposure.
3. The vaccine must contain fully active pathogens capable of producing severe disease.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Vaccination can generate memory cells. On later exposure to the same antigen, these cells allow a faster and larger antibody response.

A vaccine does not have to contain fully active disease-causing pathogens. Therefore **1 and 2 only**.

B-19

Antibiotic resistance

Answer **E**

QUESTION 19

A bacterial population contains a small number of bacteria that are resistant to antibiotic X. Antibiotic X is then used.

Which statements correctly describe natural selection in this population?

1. Resistant bacteria may have been present before the antibiotic was applied.
2. Susceptible bacteria are more likely to die, increasing the relative reproductive success of resistant bacteria.
3. The antibiotic deliberately causes every surviving bacterium to mutate to become resistant.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Resistance-conferring variation can exist before treatment. The antibiotic then acts as a selective pressure: susceptible bacteria die more readily while resistant bacteria survive and reproduce.

The antibiotic does not deliberately cause every survivor to mutate. Therefore **1 and 2 only**.

B-20

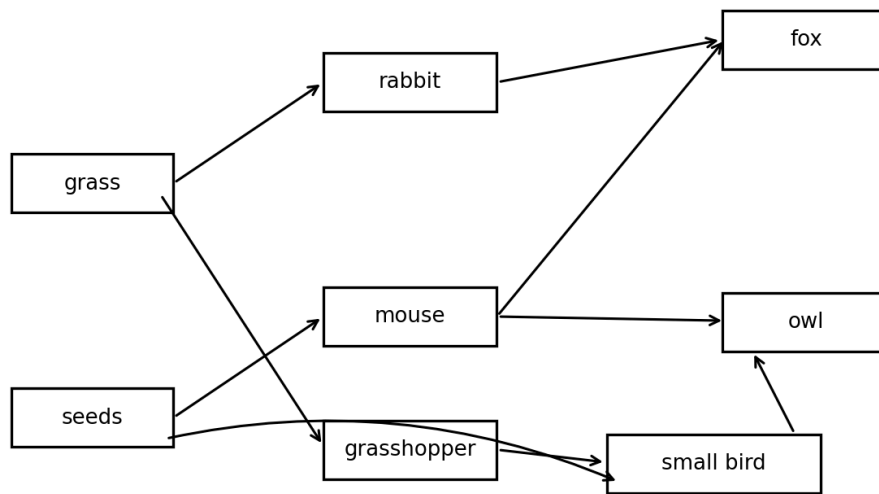
Food webs

Answer **E**

QUESTION 20

Arrows in the food web show transfer of biomass from food to consumer.

Arrows show transfer of biomass from food to consumer



Which organism can act as a **primary consumer** and a **secondary consumer**, and is also eaten by an owl?

- A** rabbit
- B** mouse
- C** fox
- D** grasshopper
- E** small bird

WORKED SOLUTION

The small bird eats seeds, so when doing so it is a primary consumer. It also eats grasshoppers, so it can act as a secondary consumer.

The food web also shows the small bird being eaten by the owl. Therefore the answer is **small bird**.

QUESTION 21

Five 1.0 m^2 quadrats are placed randomly in a 200 m^2 field.

The number of individuals of plant species X recorded is:

quadrat	number of plants
1	4
2	6
3	5
4	3
5	7

Using these samples, what is the best estimate of the number of plants of species X in the whole field?

A 200

B 400

C 500

D 1000

E 5000

WORKED SOLUTION

The mean number per 1.0 m^2 quadrat is:

$$(4 + 6 + 5 + 3 + 7) / 5 = 25 / 5 = 5$$

Estimated density = 5 plants m^{-2} .

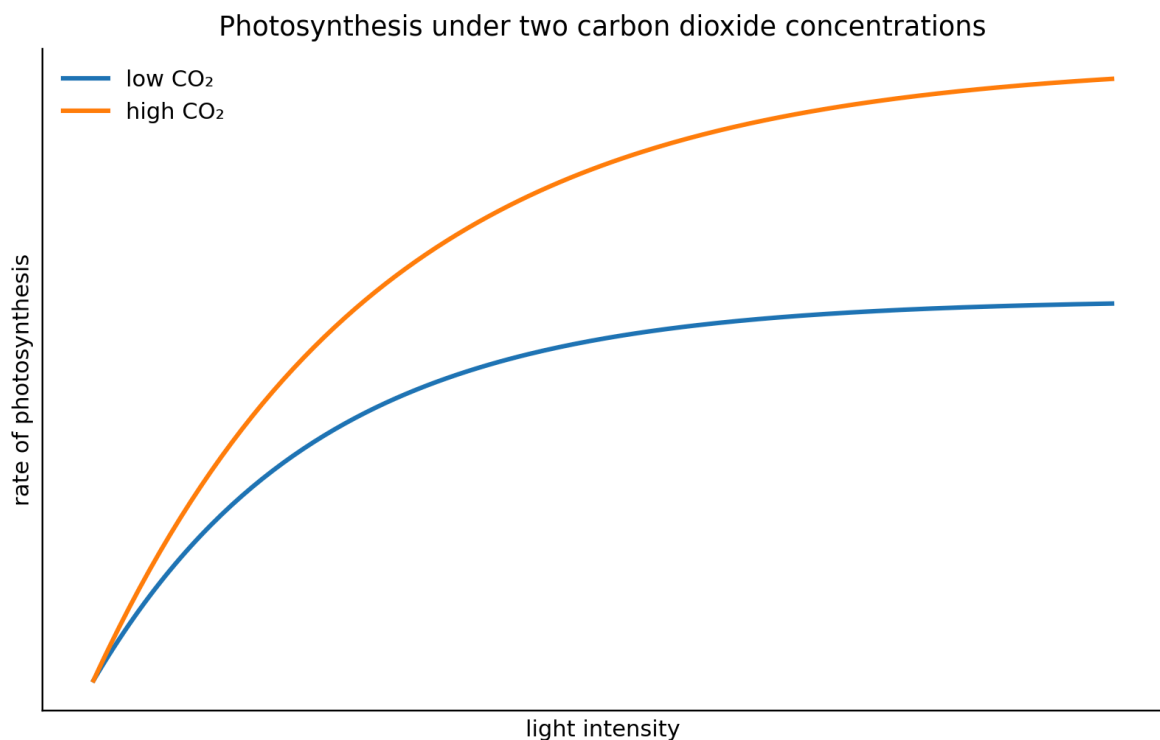
$$5 \times 200 = 1000 \text{ plants}$$

B-22

Limiting factors in photosynthesis

Answer A**QUESTION 22**

The graph shows the effect of light intensity under two carbon dioxide concentrations.



Which conclusion is best supported?

- A** At high light intensity, carbon dioxide concentration can limit the rate of photosynthesis.
- B** At low carbon dioxide concentration, light intensity can never be limiting.
- C** Increasing light intensity indefinitely must always increase photosynthesis.
- D** Carbon dioxide concentration has no effect on photosynthesis.

- E** Photosynthesis is independent of light intensity.

WORKED SOLUTION

At high light intensity, the low-CO₂ curve levels off below the high-CO₂ curve. Increasing carbon dioxide therefore increases photosynthesis under those conditions.

This supports the conclusion that carbon dioxide concentration can be a limiting factor at high light intensity.

B-23

Plant transport

Answer **B**

QUESTION 23

Which row correctly describes xylem and phloem?

- A** xylem transports dissolved sugars from leaves; phloem transports water to leaves
- B** xylem transports water and mineral ions mainly upwards; phloem transports dissolved sugars from leaves to other parts of the plant
- C** xylem transports sugars only downwards; phloem transports water only upwards
- D** xylem consists entirely of living cells; phloem consists entirely of dead cells
- E** xylem transports water by active transport along the vessel; phloem transports only mineral ions

WORKED SOLUTION

Xylem transports water and mineral ions mainly from the roots upwards through the plant.

Phloem transports dissolved sugars produced in leaves to other parts of the plant where they are used or stored.

B-24

Transpiration

Answer C

QUESTION 24

Identical leafy shoots are placed under different environmental conditions.

experiment	temperature	humidity	air movement
A	low	high	still
B	high	low	still
C	high	low	windy
D	low	high	windy

Which experiment would normally be expected to have the **highest rate of transpiration**?

A A

B B

C C

D D

E All four should have equal rates

WORKED SOLUTION

Transpiration generally increases with higher temperature, lower humidity and greater air movement.

Experiment C combines all three: **high temperature, low humidity and wind**. Therefore C should have the highest transpiration rate.

B-25

Human reproduction

Answer **B**

QUESTION 25

Which row correctly identifies the usual site of fertilisation and the usual site at which an embryo implants?

A fertilisation: ovary; implantation: oviduct

B fertilisation: oviduct; implantation: uterus

C fertilisation: uterus; implantation: ovary

D fertilisation: vagina; implantation: uterus

E fertilisation: uterus; implantation: oviduct

WORKED SOLUTION

In humans, fertilisation normally occurs in an **oviduct**. The developing embryo then reaches the uterus and implants in the **uterine lining**.

B-26

Variation

Answer **H**

QUESTION 26

Which statements are correct?

1. Human height is an example of continuous variation.
2. Human ABO blood group is an example of discontinuous variation.

3. Environmental factors such as nutrition can contribute to variation in human height.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Human height can take many values across a range, so it is continuous variation. ABO blood groups form distinct categories and are therefore discontinuous.

Height is influenced by both genetic and environmental factors, including nutrition. Therefore all three statements are correct.

B-27

Competition

Answer **B**

QUESTION 27

Two plant species are grown either separately or together for the same length of time. Assume equal starting biomasses of each species are used in the relevant treatments, and all other environmental conditions are kept the same.

treatment	mean final biomass of species A / g	mean final biomass of species B / g
A grown alone	24	—
B grown alone	—	20
A and B grown together	14	12

Which conclusion is best supported by these data?

- A** Species A and B do not compete.
- B** Growing together reduces the growth of both species, which is consistent with competition between them.
- C** Species A competes only for light.
- D** Species B will always become extinct when grown with A.
- E** The experiment proves that water is the resource for which the species compete.

WORKED SOLUTION

When grown alone, species A reaches 24 g and species B reaches 20 g. When grown together, both have lower final biomass: 14 g and 12 g respectively.

This is consistent with interspecific competition. However, the data do not identify the particular resource being competed for and do not show that either species must become extinct.

Compilation record: Level 0 uses Engineering Mock Test 1 for Mathematics/Physics and the corresponding Easy Biology/Chemistry source sets.

This solution book is ready to support the later construction of the matching full test.